

Challenge (Habanero)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** A builder is designing a house with a roof that has a 30° angle. What is the slope of a line perpendicular to the roof?
(A) $\frac{1}{\sqrt{3}}$
(B) $-\frac{1}{\sqrt{3}}$
(C) $\sqrt{3}$
(D) $-\sqrt{3}$
(E) 0
- Q3.** A recipe for making cookies calls for a 3 : 5 ratio of sugar to flour. If a line represents the amount of sugar needed, what is the slope of a line perpendicular to this ratio?
(A) $-\frac{5}{3}$
(B) $\frac{5}{3}$
(C) $-\frac{3}{5}$
(D) $\frac{3}{5}$
(E) 0
- Q4.** What is the equation of a line perpendicular to $y = -3x - 2$ and passing through (1, 4)?
(A) $y = \frac{1}{3}x + \frac{11}{3}$
(B) $y = -\frac{1}{3}x + \frac{11}{3}$
(C) $y = \frac{1}{3}x - \frac{11}{3}$
(D) $y = -\frac{1}{3}x - \frac{11}{3}$
(E) $y = \frac{1}{3}x + \frac{5}{3}$
- Q5.** A contractor is building a staircase with a 45° angle. What is the slope of a line perpendicular to the staircase?
(A) 1
(B) -1
(C) 0
(D) $\frac{1}{2}$
(E) $-\frac{1}{2}$
- Q6.** A chef is preparing a dish that requires a 2 : 7 ratio of oil to vinegar. If a line represents the amount of oil needed, what is the slope of a line perpendicular to this ratio?
(A) $-\frac{7}{2}$
(B) $\frac{7}{2}$
(C) $-\frac{2}{7}$
(D) $\frac{2}{7}$
(E) 0
- Q7.** What is the equation of a line perpendicular to $y = \frac{2}{3}x - 1$ and passing through (2, 5)?
(A) $y = -\frac{3}{2}x + 8$
(B) $y = \frac{3}{2}x - 8$
(C) $y = -\frac{3}{2}x + 2$
(D) $y = \frac{3}{2}x + 2$
(E) $y = -\frac{3}{2}x + 1$
- Q8.** A carpenter is building a shed with a roof that has a 60° angle. What is the slope of a line perpendicular to the roof?
(A) $-\frac{1}{\sqrt{3}}$
(B) $\frac{1}{\sqrt{3}}$
(C) $\sqrt{3}$
(D) $-\sqrt{3}$
(E) 0

Open Ended Questions (FRQ)

- Q9.** A architect is designing a building with two perpendicular walls. One wall has a height of 20 meters and the other wall has a height of 15 meters. If the walls are perpendicular, what is the equation of the line that represents the relationship between the heights of the walls? Show your work and explain your answer.
- Q10.** A recipe for making bread calls for a 2 : 5 ratio of yeast to flour. If a line represents the amount of yeast needed, what is the equation of a line perpendicular to this ratio? Show your work and explain your answer.

Chef's Tips

- Tip 1: Make sure to check the slopes of perpendicular lines carefully.
- Tip 2: Use the negative reciprocal to find the slope of a perpendicular line.
- Tip 3: Always show your work and explain your answers clearly.